Python Quick Start Guide for Windows: Mech. Actuators Traner

STEP 1 Check Components and Details

Make sure your Mechatronic Actuators Trainer case includes the following components.

1. Mechatronic Actuators Trainer
2. 4 Motors (1x DC, 1x BLDC, 1x Stepper, 1x Servo)
3. Power Supply & Cord
4. USB C Cable
5. Metal Motor Hub Attachments
6. Servo Horn Set
7. Extra connectors for External Motors
8. Actuators Trainer Resources. Content and courseware provided digitally at github.com/quanser/Quanser_Academic_Resources

Ensure you local computer has the following components

1. Windows 10/11 operating system
2. Quanser SDK 2026 or QUARC 2026 or later for either, installed

STEP 2 Download Content and Set Up Computer

A

Download the resources for the device from
https://github.com/quanser/Quanser_Academic_Resources
Follow the instructions on the 'Downloading Resources' and 'Setting Up Your Computer' sections.

B

You should now have a supported Python version downloaded and added to PATH and the 'Completing the Setup' section from 'Setting Up Your Computer' should have required a restart of your PC after installing all necessary libraries.

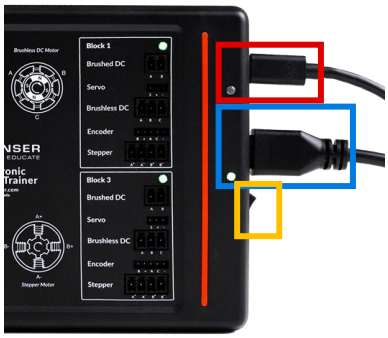
C

Download VS Code or a code editor of your choice. If you use VS Code, please install the Python extension (VS code will suggest it once you open the python file in step 4). You should now have all the necessary files to run the device.

STEP 3 Set Up the Hardware

The steps below outline the instructions to setup the Mechatronic Actuators Trainer.

A



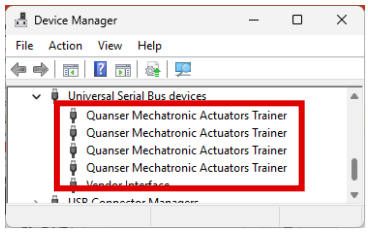
First, connect the supplied 12V power supply to the Power connector on the Actuators Trainer and then to a wall outlet using the included power cable.

Next, connect the USB C connector to the device and then to an enabled USB 3.0 (or higher) port on your desktop PC or laptop.

Move the Power Switch to the ON position. Both Power and USB LED should turn green.

If step 2 was done successfully and QUARC or Quanser SDK were properly installed, Windows[1] should automatically detect the presence of the Actuators Trainer.

B



Go to Windows Device Manager and verify that the device appears 4 times in the list under *Universal Serial Bus devices*. Note that the computer should recognize each Block of the device as a different device, explaining the 4 instances in the list.

STEP 4 Testing the Mechatronic Actuators Trainer

Follow the procedure below to test your Actuators Trainer.

A

Connect the motors as shown in these pictures. Each motor can be connected to any block.

Note that the empty cable in the encoder connector for the DC motor (2) should match the C channel on the board. If you connect the encoder cable of the BLDC instead, connect the black wire to -.

Note that the orange wire in the servo connector (3) should match the S connector on the board.

B

Open **`quick_start_actuators.py`** on your code editor (VS Code) and click run.

C

1. This code is meant to run all motors, no matter what block they are connected to and print the encoder counts of the motor that is connected to the encoder port. Test all ports by moving the motors.
2. Note that if BLDC is connected to the encoder, the counts are around 360 cps, for DC its around 130k. This will be printed to the terminal every 2 seconds.
3. The code automatically stops after 200 seconds, or you can stop it using Ctrl+C on the terminal.

```
/2_quick_start_guides/mech_actuators_trainer/quick_start_actuators.py"
motor speed block 0: 0.00 counts/s    motor speed block 1: 133004.93 counts/s    motor speed block 2: 351.84 counts/s    motor speed block 3: 0.00 counts/s
motor speed block 0: 0.00 counts/s    motor speed block 1: 132271.89 counts/s    motor speed block 2: 363.79 counts/s    motor speed block 3: 0.00 counts/s
KeyboardInterrupt handled. Cleaning up.
Actuators Trainer Block 3 closed gracefully.
Actuators Trainer Block 2 closed gracefully.
Actuators Trainer Block 1 closed gracefully.
Actuators Trainer Block 0 closed gracefully.
```

TROUBLESHOOTING

Review the following recommendations before contacting Quanser's technical support engineers.

1. Check the connections outlined in Step 3 of this guide and ensure that the cables are connected firmly.

STILL NEED HELP For further assistance from a Quanser engineer, contact us at tech@quanser.com

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[1] Windows is a registered trademark of Microsoft Corporation in the United States and other countries.